PSTR047.02

The effect of reward on existing neutral memory representations and consequences for decision-making

Bakkour Memory and Decision Lab

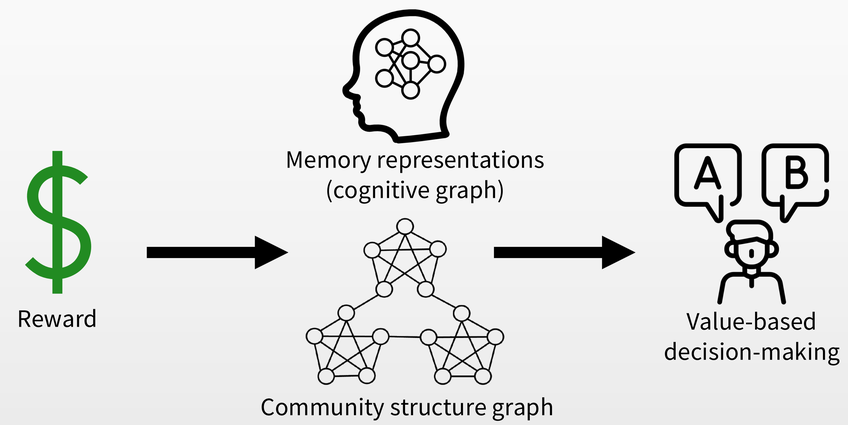
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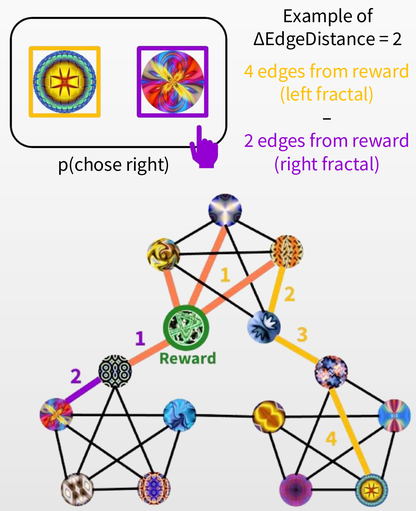
seoyounglee@uchicago.edu

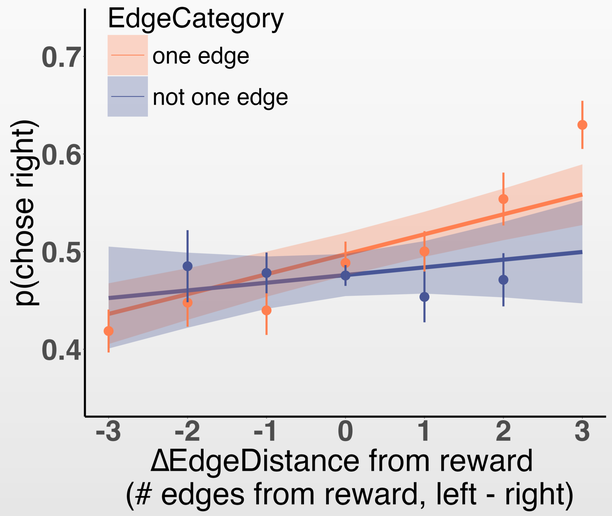
How does reward affect memory representations and decision-making?

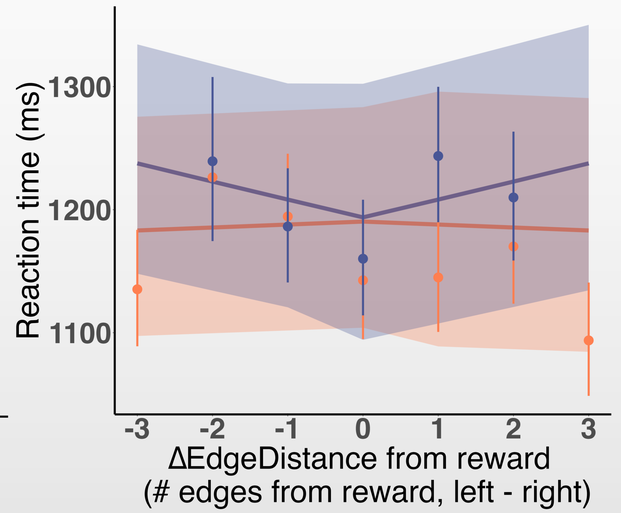


1) *Does reward spread in a graded manner throughout a community structure cognitive graph and affect decision-making?*
2) *Does the community structure affect value-based decision-making?*

Choices were driven by distance from reward (only for trials including fractals that are one edge from reward) and decision times were *not* driven by distance from reward

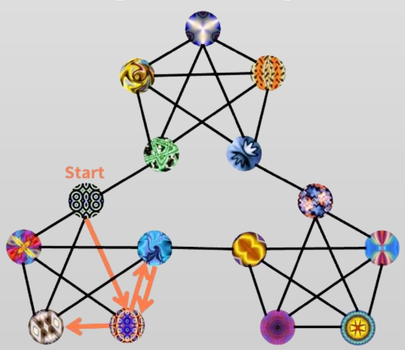




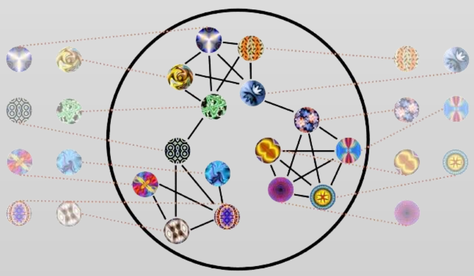


Methods

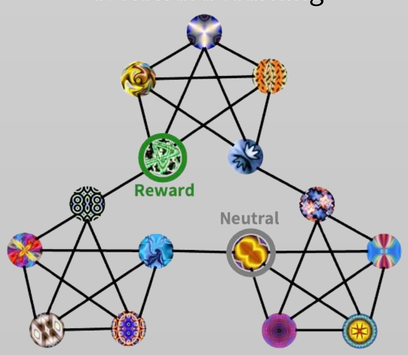
1. Community structure graph learning



2. Graph test

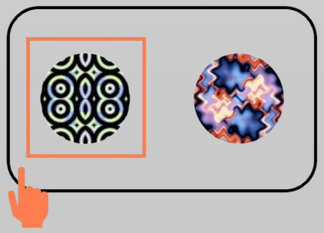


3. Reward learning

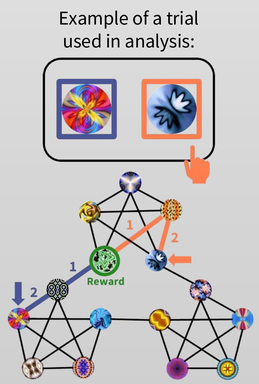


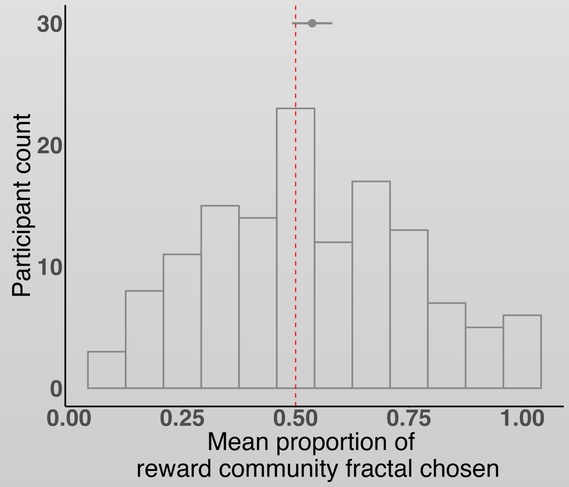
4. Decision-making

Choose which fractal is more likely to lead to monetary reward:

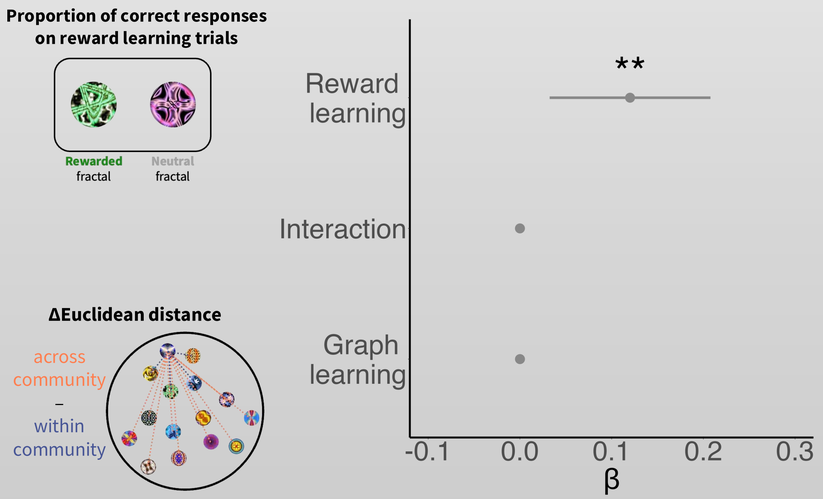


No effect of community structure on choices



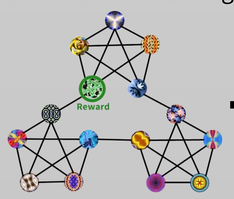


Performance on reward learning affects choice more than performance on graph learning



Future directions: Has the community structure been reorganized?

Before reward learning:



After reward learning and during decision-making:



Conclusion

Reward spread throughout an existing neutral community structure cognitive graph and affected value-based decision-making.